Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/02/2010
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Project Title:	Determination of the Bone Marrow Macrophage Phenotype in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
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Zip Code:	66506-4901	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:	Shuttle/ISS		
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The opportunity to fly bone marrow cells from C57BL/6 mice in space flight would allow the Chapes lab to test hypotheses related to those proposed in the original Bonemac project (NAG2-1274) which was cancelled before completion. The Bonemac project was not completed because of hardware considerations, the Columbia space shuttle accident, and NASA's change in priorities. The ULF-2 mission on STS-126 will provide the opportunity to test the hypothesis that the expression of the macrophage colony stimulating factor receptor (c-fms) or the expression of c-fos on bone marrow cells differentiating in the presence of macrophage stimulating factor (CSF-1) while those cells are flown in space. This will assess the mechanism of how space flight affects the differentiation of those bone marrow cells into macrophages by determining if these critical elements are altered in the cells as they differentiate.
The study will also complement recently completed experiments that were part of the CBTM-2 payload that flew on STS-118. We discovered that there were significant impacts on cells differentiating in the granulocytic lineage,

Task Description:	including macrophages. Therefore, STS-118 provided additional evidence that space flight affects hematopoiesis of cells in the granulocytic lineage, including macrophages. Although the ULF2 mission will fly isolated bone marrow ex vivo, it will allow us to assess the effects of space flight on bone marrow cells directly. It will provide the opportunity to determine at what critical stage the macrophage differentiation process is disrupted. By measuring key changes in gene expression and cell surface phenotypes of cells as they differentiate we will be able to determine where the process is affected. We will assess macrophage differentiation at three different time points during the flight. In particular, we will do temporal comparisons of the expression of several different cell surface markers which distinguish phenotypically distinct subsets in the bone marrow as well as mature and immature macrophages, neutrophils, and other cells. After the STS-126 flight, cells will be recovered, probed with antibody, and analyzed using flow cytometric techniques. Some cells will be preserved in RNA-Later. These cells will be recovered, treated with Trizol, and RNA will be used in an expression analysis of various signaling genes involved in the differentiation responses. Some samples will not be fixed and will be used to assess cell viability and the supernatants will be analyzed for cytokines. These studies will help identify how the human body adapts to space flight and identify countermeasures to these changes.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space flight offers a unique environment that impacts many physiological systems. Amongst these systems, bone metabolism is known to be significantly altered by the skeletal unloading of space flight. Hematopoiesis is the process which provides the body with mature, functioning cells necessary for oxygenation, clotting and immune functions. The bone, specifically, the bone marrow, is the site for hematopoiesis. Moreover, changes in bone impact all compartments therein, including the marrow. Indeed, analysis of animals from rats to rhesus moneky to humans shows that hematopoiesis can be affected by space flight. Although direct examination of primary hematopoiesis is not an aim of the proposed project, this experiment will allow us to dissect a very complex system. We will obtain valuable information about what mechanisms involved in hematopoiesis are susceptible to disruption by space flight. This understanding will allow for more effective countermeasure planning. On Earth, skeletal unloading occurs to individuals that are bedridden for long periods of time, wheel-chair bound, or lose the ability to stand for long periods of time. These prolonged periods of skeletal unloading also induce changes in bone that parallel those seen in space flight. In addition, individuals with bone disorders, such as osteoporosis, are also subject to alterations in hematopoiesis. This is best exemplified by the osteopetrotic mouse, which has diminished numbers of macrophages and brittle bones. Therefore, data obtained from this study will also have direct extrapolations to individuals on Earth.
Task Progress:	On November 14, 2008, the BoneMac experiment was launched on STS-126/ULF2 (Endeavour). The experiment was conducted on the Middeck. All in-flight experiment operations were completed per schedule. Endeavour returned to Earth on November 20, 2008, with the landing occurring at the Edwards Air Force Base alternate landing site. The BoneMac specimens were flown to Kennedy Space Center Space Life Sciences (SLS) Lab via a commercial flight that departed from Los Angeles on the Orbiter landing date. The total mission duration was 15 days. The goal of this flight research experiment was to determine how macrophage differentiation is suppressed in the space flight environment. The specimen used for this study were primary murine (mouse) bone marrow cells. The specific objectives were as follows: • Test that hypothesis progenitor macrophage differentiation is disrupted by CSF-1 receptor expression down regulation . • Assess the differentiation mechanism by measuring how space flight affects the differentiation of bone marrow cells into macrophages by cell phenotying, assessing mRNA expression, and assessing cytokine production by the bone marrow cells during macrophage progenitor cell differentiation. • Analyze changes in cell surface phenotypes and gene expression during differentiation to identify where the differentiation process is affected by the space flight environment. Primary murine bone marrow cells from adult C57BL/6J mice were used for this experiment. The bone marrow cells were collected from the humeri, femora, and tibiae. The collected cells were suspended in growth medium that was supplemented with macrophage colony stimulating factor (CSF-1). The cells were loaded into the primary chamber of 48 FPAs. In 32 FPAs, formalin was loaded into the secondary chamber. Eight FPAs were loaded with 6.0 M guanidinium isothiocyanate (GITC) in the secondary chamber, and, 8 FPAs were loaded with growth medium plus CSF-1 (nonpreserved) in the secondary chamber. A total of 6 GAPs were loaded with 8 FPAs each. After loading, the cells were placed into the CGBA (37 oC) to start incubation, pre-launch. After launch, BoneMac GAP crew operations were conducted on two dates. One set of GAPs were operated on FD6, and a second set of GAPs were operated on FD15. This timeline allowed for obtaining bone marrow cells that were allowed to differentiate for different time durations. Post-flight, the FPAs were returned to the KSU laboratory for processing and analyses. A ground control was run at SLS Lab using FPAs, GAPs, and a CGBA. This ground control followed the same protocols and timeline as the flight samples. We determined that the nonpreserved (kept in medium) set had more cells in the flight FPAs (n=7; 3.0±0.6 x 107; mean±Sd) compared to the ground (n=4; 1.7±0.4 x 107; P<0.01, t-test; mean±Sd). We have also found there were more cells in the formalin-fixed flight FPAs 5.3±0.6; n=28; mean±Sd) compared to the ground (4.4±0.6 x 106; n=32; mean±Sd; P<0.01, t-test). The increase in cell number from the time the cells were fixed on Day 14 of flight to landing and recovery on Day 17 suggests that the cells were viable at the time of fixation. Therefore, there appears to be more proliferation in space than on the ground with flight cell numbers increasing 5.66 fold and ground cell numbers increasing 3.86 fold from day 14 to day 17. We also measured glucose utilization by the cells in the unfixed FPA's. We found a significantly less glucose usage by flight cells (121±4 mg/dl) compared to cells grown on the ground (159±3 mg/dl). These data suggest that the cells required less energy to proliferate more in space. We found a significant increase in the overall expression of Mac2+,c-fms+ cells and Mac2+, c-fos+ cells differentiated in space compared to ground controls and a significant change in the distribution of cells that were double positive for Mac2 and c-fms and c-fos between the two treatment groups (P<0.05, chi 2 analysis). However, we did not see a change in distribution in the expression of F4/80+,CD44+ cells, F4/80+,Ly6C+ cells or F4/80+,Gr1+ cells. The increased level

	of expression of F4/80 and Mac2 and the decreased expression of Ly6C suggests that space flight may be increasing driving cells toward being more differentiated. The increased expression of c-fms, especially in the populations that would be particularly rich in differentiated macrophages would be consistent with the idea that binding more macrophage colony stimulating factor may drive the cells to become more differentiated. We also reject the hypothesis that space flight decreases the expression of c-fms, the receptor for CSF-1. In contrast, it appears to increase its expression, especially on the Mac2+, c-fms+ cell populations By 15 days of flight the cells in the FPA should have been differentiated into the macrophage lineage. For flight samples, we found that 209 genes had gene transcript levels >1.5 fold higher than ground controls. In contrast, we found that 644 genes had gene transcript levels >1.5 fold lower than ground controls. The genes were broadly categorized into 16 broad categories based on function. These included: Genes involved in carbohydrate synthesis, cell division, cytokine/hormone function, cytokine/hormone receptor, hematopoiesis, host defense, lipid synthesis, protein synthesis, nucleotide synthesis, signal transduction, structural protein, transcription, translation, transport, Energy or metabolic processes, Development, or protein function. Genes of unknown function 291 genes of unknown function were also identified. Although the analysis of this array will continue to be mined, the data obtained show that this part of the experiment was successful.
Bibliography Type:	Description: (Last Updated: 09/15/2021)
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